

Extra Credit Project 2

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Project Goal:

The goal of the project is to implement an exact solver for the Traveling Salesman Problem (TSP) using Dynamic Programming. While TSP is NP-hard, this project aims to demonstrate that through memoization and state-space reduction, we can find the absolute optimal route for a meaningful number of real-world locations far more efficiently than a brute-force search.

Algorithm Description:

The core of the project is the Held-Karp algorithm. Instead of checking all permutations, we use the principle of optimality to solve sub-problems.

The Held-Karp algorithm uses memoization to solve the TSP by breaking it into overlapping sub-problems. Guaranteed to find the absolute shortest path. Because of the exponential complexity, this will be restricted to a smaller subset of the data.

Dataset and size: I plan to use latitude and longitude coordinates for 10-15 cities for a region.

Application:

I plan to provide an application with a visual and analytical breakdown of the results:

Using Matplotlib, the application will plot the cities and the calculated optimal path to verify that the route is logical and does not cross over itself unnecessarily.

A comparison of the number of operations and time required for brute force vs. dynamic programming to highlight the power of DP.